

User Instructions/Guide

Topic: DrillMain Intelligence

Smart drilling insights with next-step guidance and integrated maintenance.

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Course: CSIS 4495 002 Applied Research Project

Section: 002

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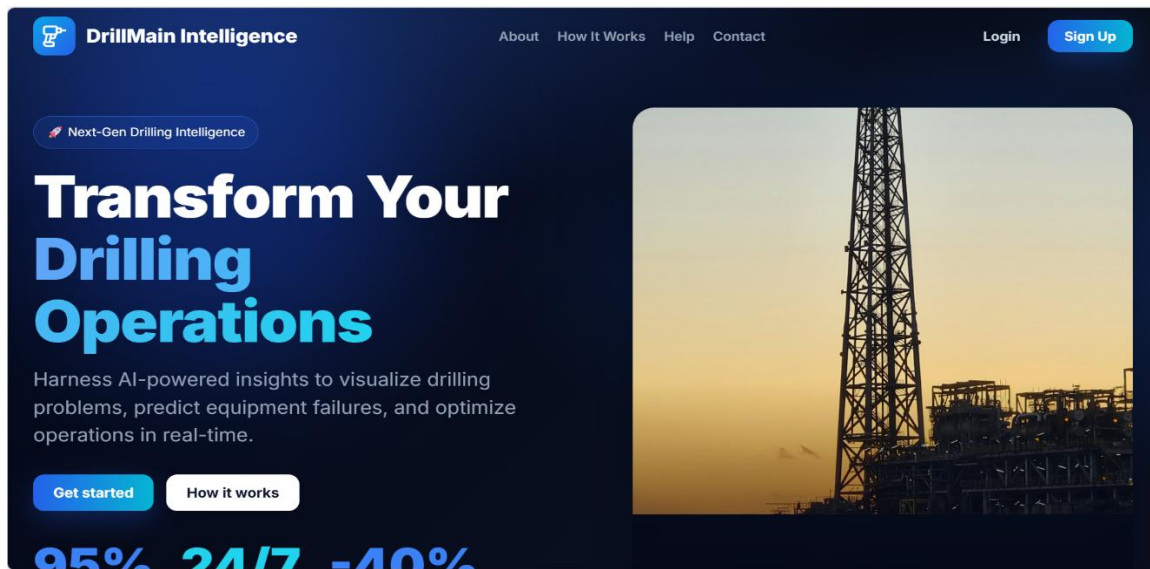
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Home Page

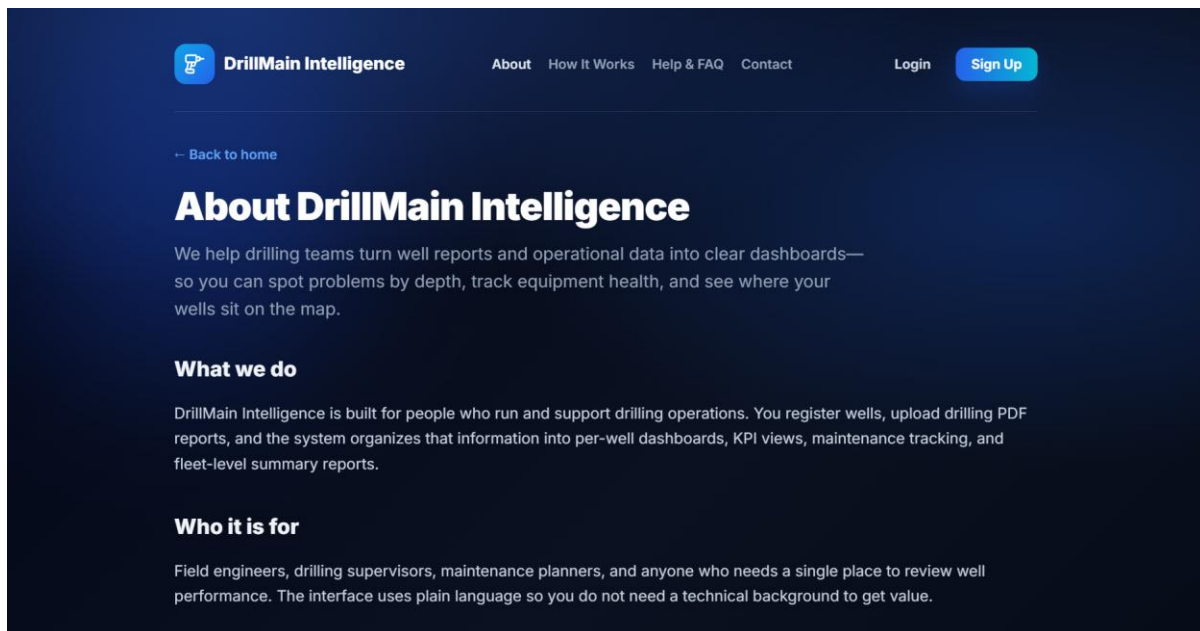
1.1 Home screen:

- The first thing you see on your screen after loading the website is the home page (see pictures below).



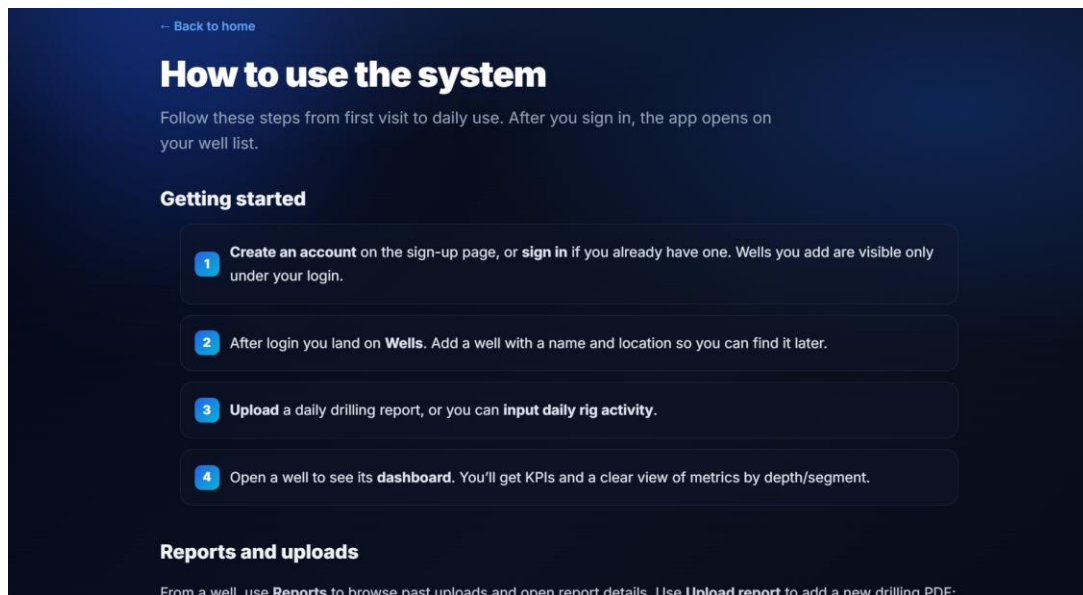
1.2 About Page:

- On the Home page, click on “About” link at the top of the page. This page explains what DrillMain Intelligence gives you a brief introduction of what the app is all about.



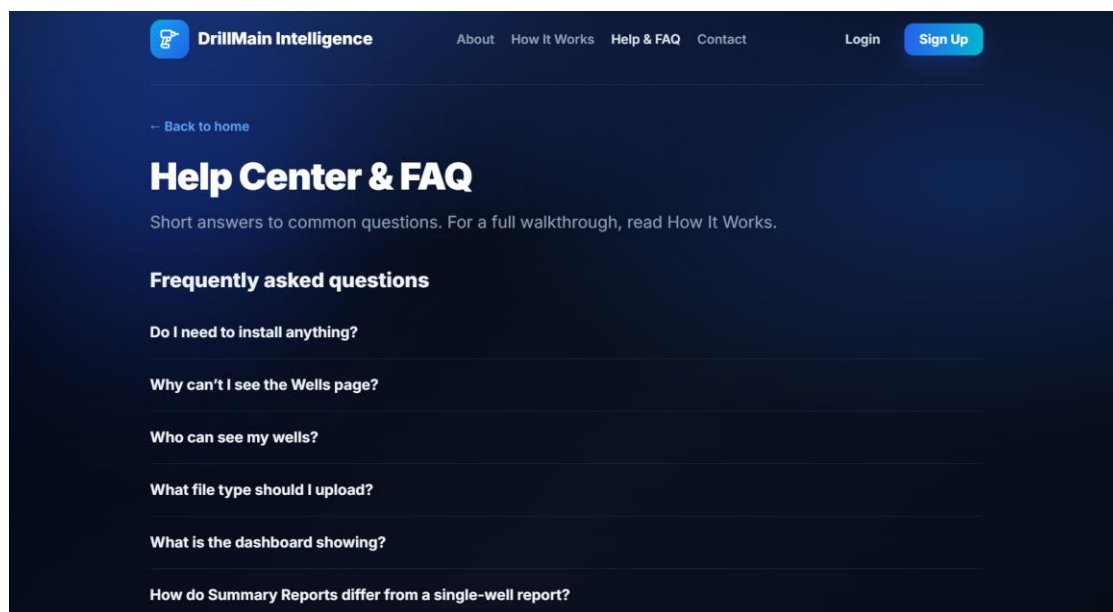
1.3 How it works:

- On the Home page, click on “How it works” link at the top of the page. This page gives a brief overview on how to use the system.



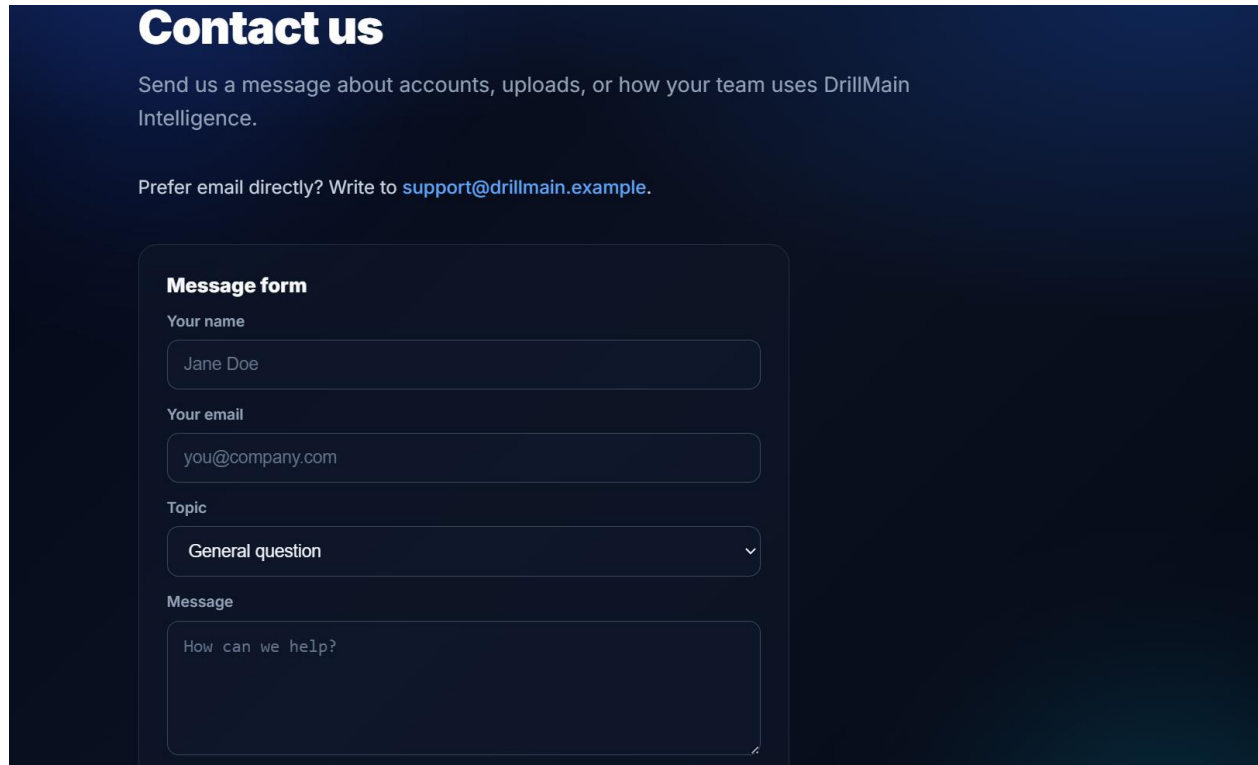
1.4 Help & FAQ:

- On the Home page, click on “About” link at the top of the page. This page explains what DrillMain Intelligence gives you a brief introduction of what the app is all about.



1.5 Contact us:

- On the Home page, click on “About” link at the top of the page. This page explains what DrillMain Intelligence gives you a brief introduction of what the app is all about.

A screenshot of a 'Contact us' form on a dark blue background. The form is titled 'Contact us' in large white text. Below the title, there is a subtitle: 'Send us a message about accounts, uploads, or how your team uses DrillMain Intelligence.' followed by a line of text: 'Prefer email directly? Write to support@drillmain.example.' The form itself is a light blue rounded rectangle with the title 'Message form' in bold. It contains four input fields: 'Your name' with the text 'Jane Doe', 'Your email' with the text 'you@company.com', 'Topic' with a dropdown menu showing 'General question' and a downward arrow, and 'Message' with the text 'How can we help?'.

Contact us

Send us a message about accounts, uploads, or how your team uses DrillMain Intelligence.

Prefer email directly? Write to support@drillmain.example.

Message form

Your name

Jane Doe

Your email

you@company.com

Topic

General question

Message

How can we help?

Registration

2.1 The sign-up page.

- On the home page, click on the sign-up button to create an account, by inputting your details.

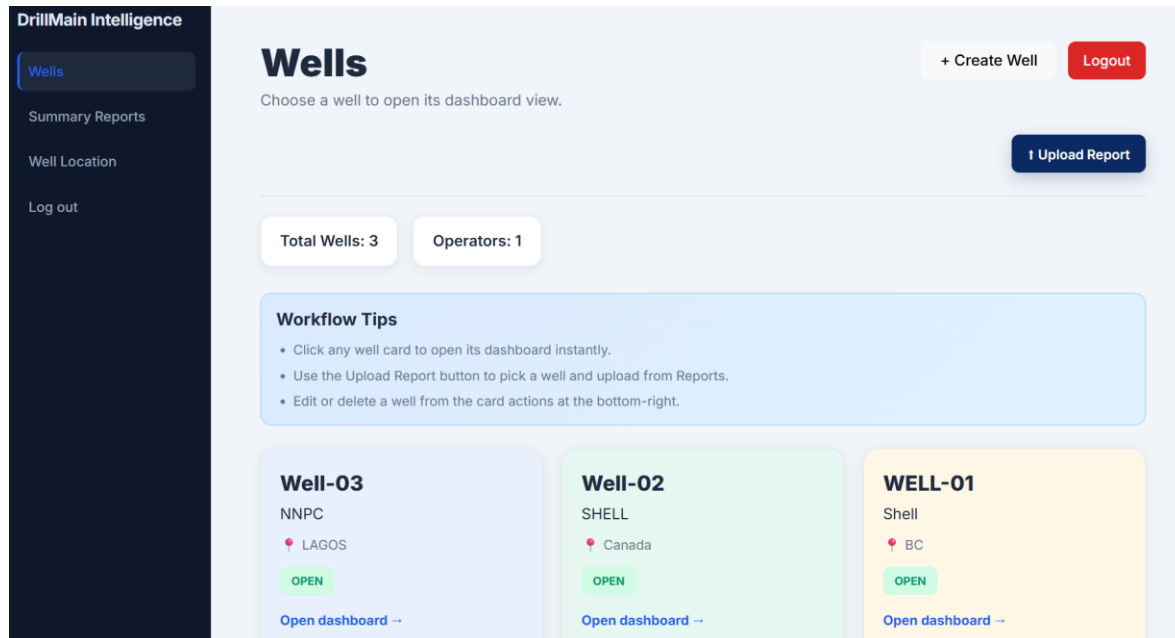
2.2 The login page

- After signing up, you will be directed to the login page.
- Fill in your login details to be able to login.

Creation of Wells

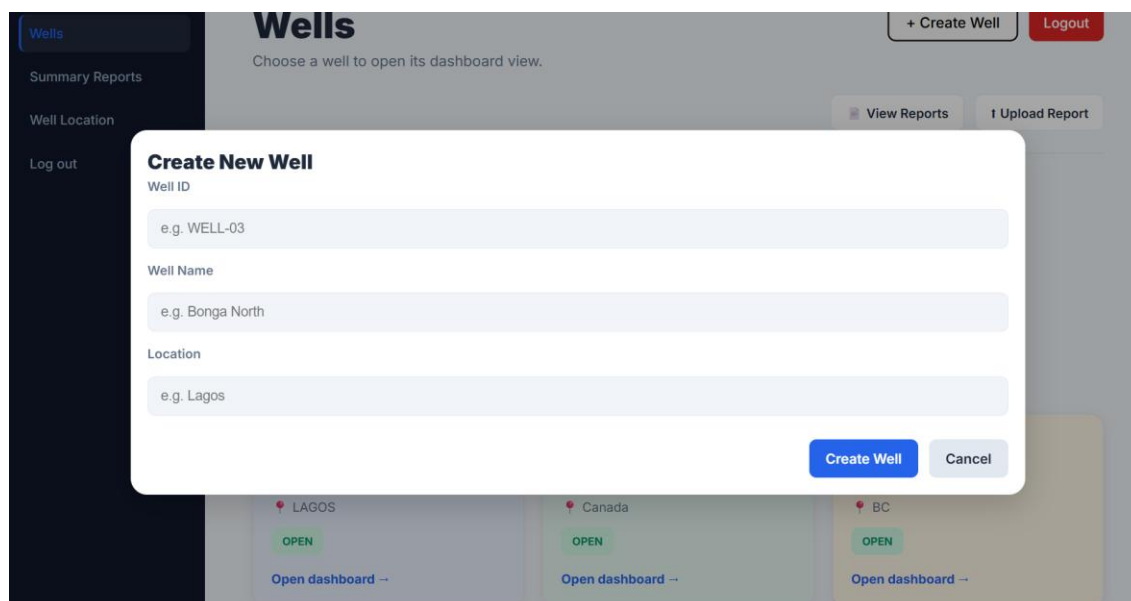
3.1 Wells Page

- After you login, you will be directed to the wells page.



3.2 Create New Well

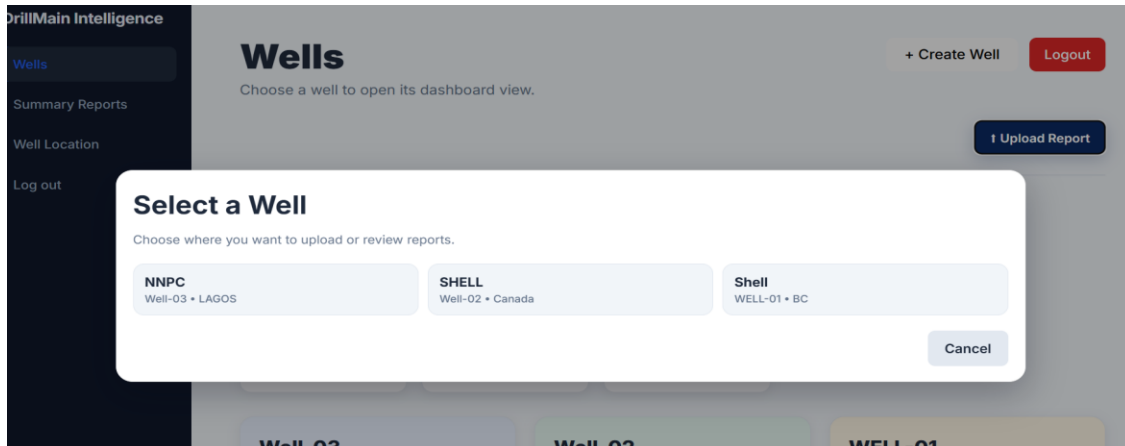
- Click on the create well button to register a new well.
- Fill the well ID, well name and location.
- Click “create well” button to create a new well.



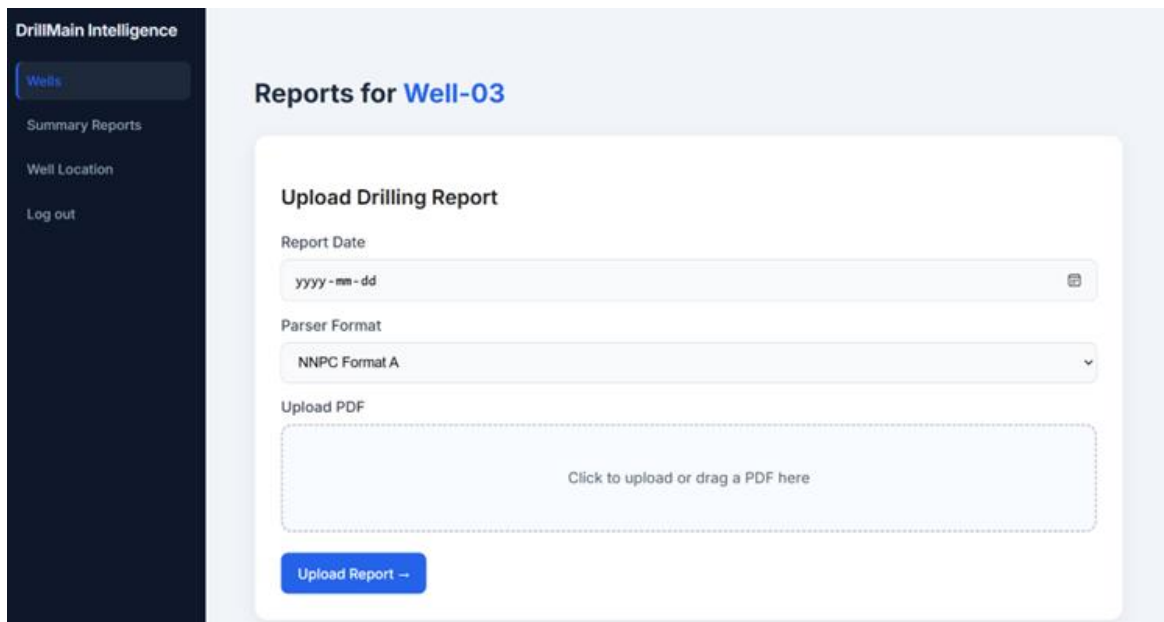
Reports

4.1 uploading Reports

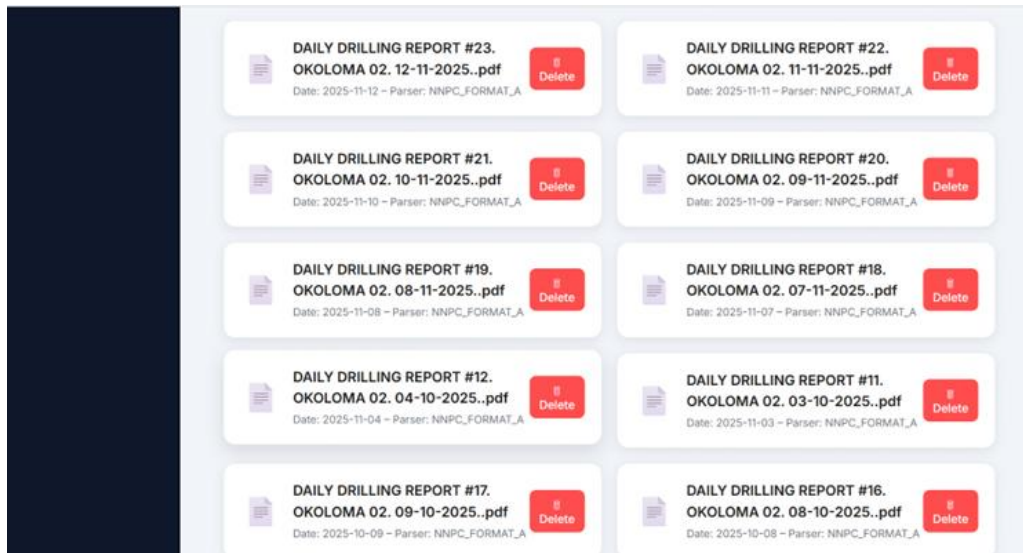
- On the wells page, click on upload report. This will open a window showing the list of wells, click on the well you would like to upload your report.



- After selecting a well, you will be directed to the reports page where you will upload your report.
- Fill in the report date choose the format and upload your pdf.



- Scroll down on the report page to see the uploaded reports.



4.2 Report Details / Editing report

- Click on the report to view the details of the report.

DrillMain Intelligence

- Wells
- Summary Reports
- Well Location
- Log out

[← Back to Reports](#)

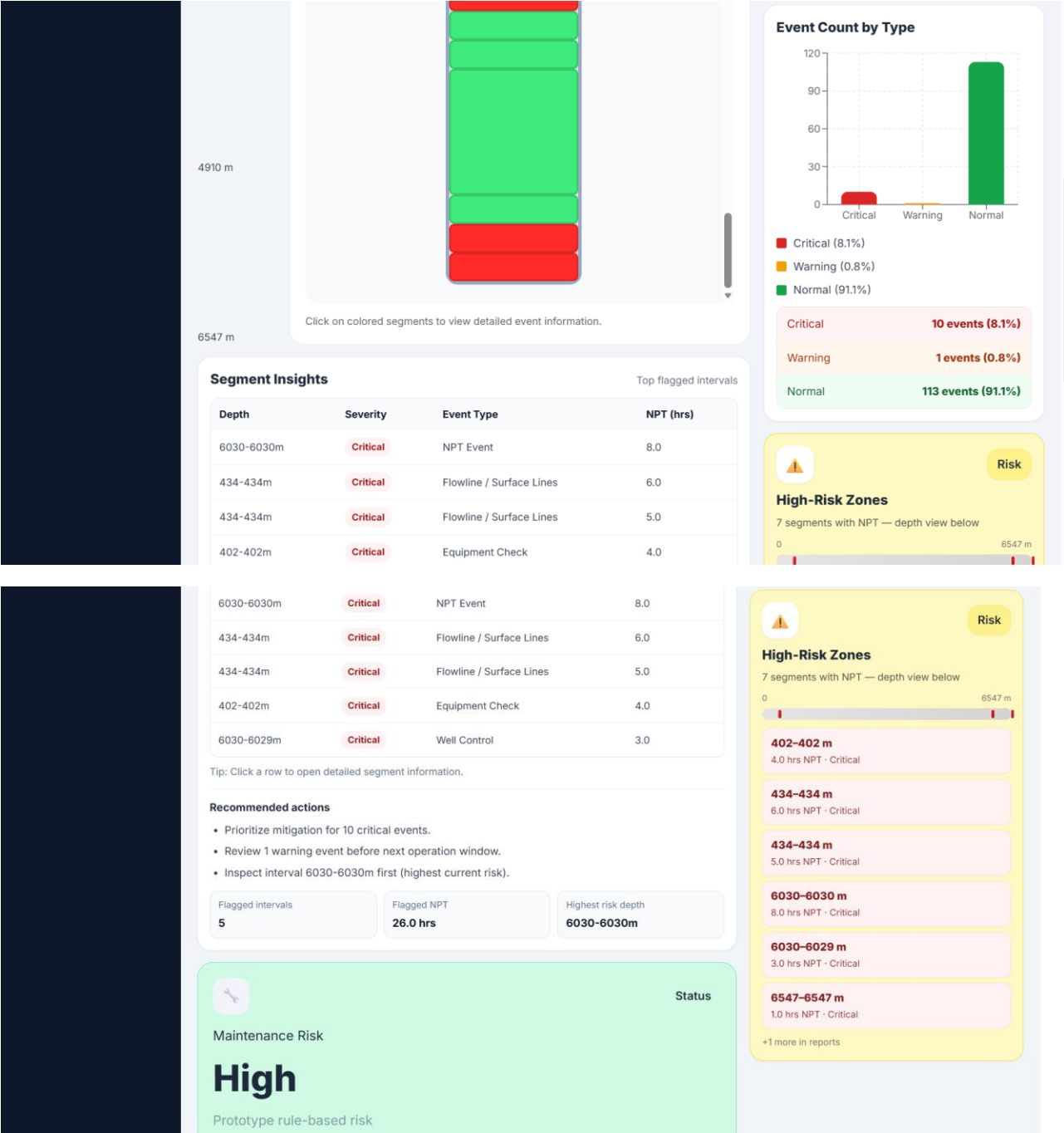
Report #20 — DAILY DRILLING REPORT #19. OKOLOMA 02. 08-11-2025..pdf

Well: Well-03
Date: 2025-11-08
Parser: NNPC_FORMAT_A

Extracted Operations

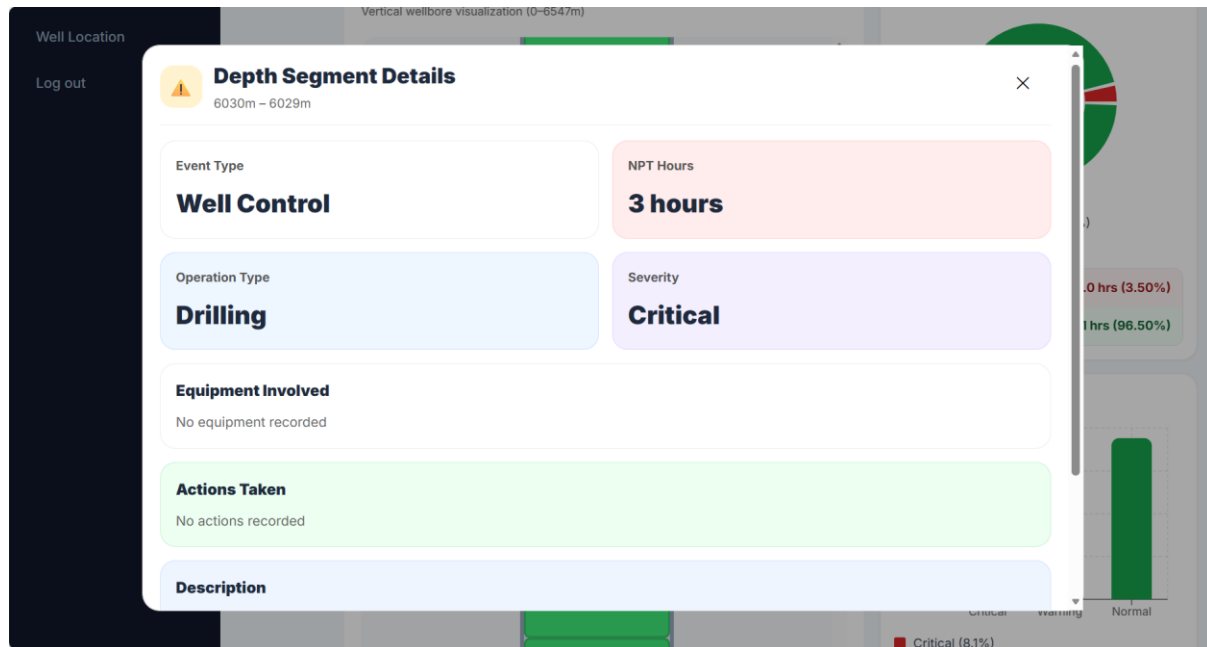
Depth From	Depth To	Operation Type	Duration	NPT	Description
6030	6030	Drilling	5.5		PREPARE THE BOP STACK IN SET BACK AREA FOR INSTALLATION. P/U AND LOWER 13-5/8" 5M x 13-5/8" 10M DSA ON 13-3/8" 5M x 13-5/8" 5M CHH AND TIGHTENED CONNECTION BOLTS. P/U & MOVE THE 13-5/8" 10M BOP STACK FROM THE SET BACK AREA & N/U SAME ON 13-5/8" 5M x 13-5/8" 10M DSA AND TIGHTENED CONNECTION BOLTS.
6030	6030	Drilling	17.5		POWERED UP KOOMEY UNIT AND FUNCTION TESTED SAME-OK. R/U DS FRONT LADDER, CATWALK & V-DOOR CONNECTED HYDRAULIC CONTROL LINE FROM KOOMEY UNIT TO BOP. R/U BELL NIPPLE AND SECURE SAME TO THE BOP.

- Click on the “edit report” button to make changes if necessary.
- After clicking on the edit report button, the edit page will open to make changes on the report or delete data.
- Click on “save changes” after making changes to the report.

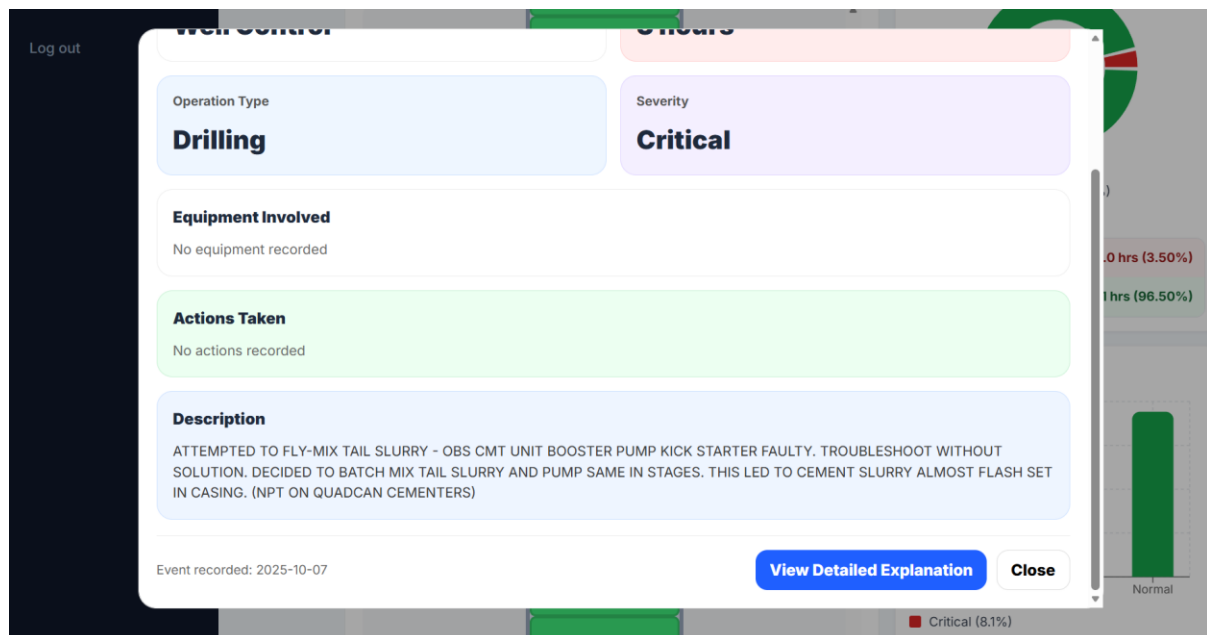


5.2 Segment Details Panel

- Click on the segment and a segment modal will pop up showing a summary of what went on at a particular depth, the severity and the type of event that occurred.



- Click on “View Detailed Explanation” to view the analysis of the event that occurred at a particular depth.



Drilling Dashboard — Well-03

Critical Event Analysis

Depth: 402.0m - 402.0m

Close

How We Determined This Title

The title was set to 'Critical Event Analysis' because the segment severity is critical and no specific event keyword (e.g. stuck pipe) was found in the description.

Why Was This Flagged?

This segment is classified as critical because high non-productive time (4.0 hours; ≥ 2 hours triggers critical classification). The contributing factors below add detail where relevant.

Contributing Factors

High NPT in This Interval

The report explicitly records 4.0 hours of non-productive time for this segment (depth 402.0m–402.0m, operation: 'Drilling'). Our rule: $\text{NPT} \geq 2$ hours classifies as critical. Elevated NPT in this depth range indicates operational issues that warrant closer review and mitigation planning.

Similar Events in Well History

Technical Factors Identified

- Formation permeability changes at depth boundary.
- Mud cake buildup in permeable zones.
- Tight clearance between drill string and wellbore.

Recommended Prevention Measures

- ✓ Optimize mud weight for overbalance in this depth range.
- ✓ Minimize static time in this interval.
- ✓ Monitor torque and drag during operations like this one.

Analysis Methodology

This analysis uses the segment's depth interval, operation type, NPT hours, and description from the report. Segment severity (wellbore color) is set by AI when available, otherwise by rule-based logic; the title and flagged explanation match that same level. Contributing factors and recommendations are derived from the report fields.

Back to Details

Close Analysis

- Click on “Back to details” to go back to the segment modal or click on “Close Analysis” to go back to the dashboard.

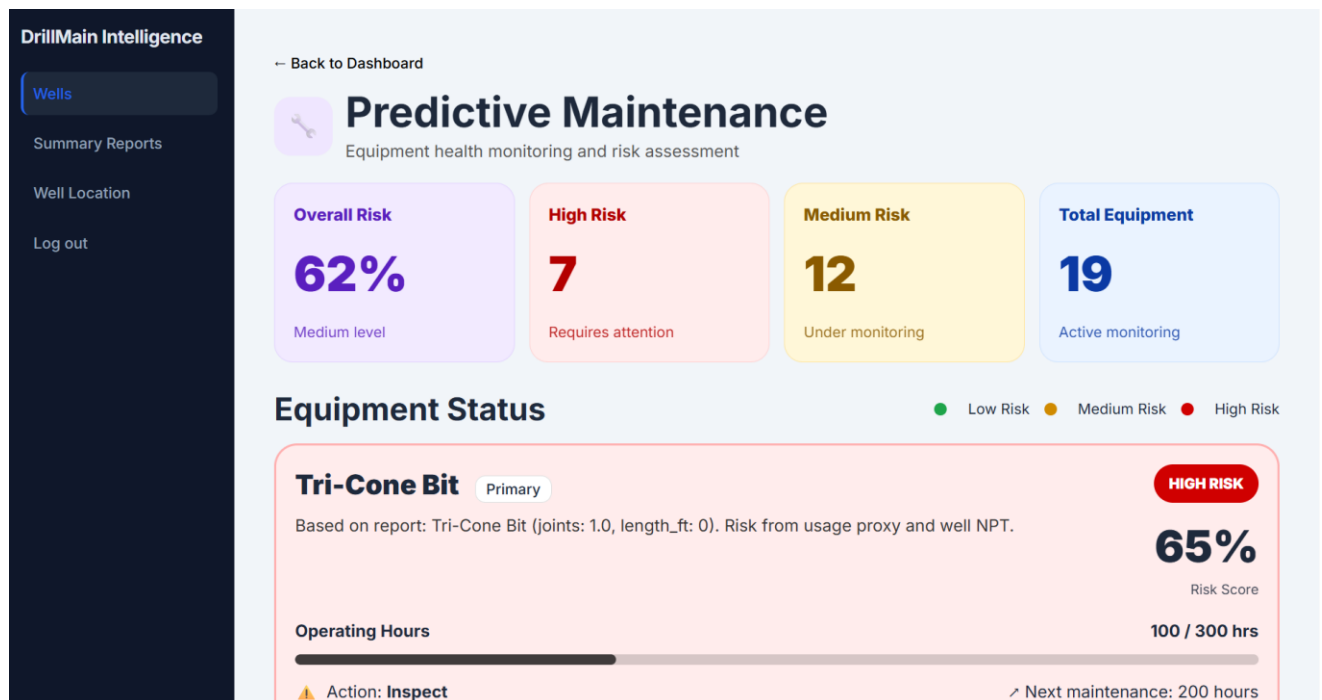
5.3 KPI Cards

- Click on the **NPT vs Productive Time** KPI card on the dashboard to view more information on the Non-productive time.
- Click on the **Event Count by Type** KPI card on the dashboard to view more information on the event count based on the type of event that occurred.

Prédicative Maintenance

6.1 Prédicative Maintenance Page

- Click the predictive maintenance button in dashboard page to open maintenance page for your equipment.



Operating Hours

237 / 2000 hrs

Action: Monitor

Next maintenance: 1763 hours

Heavy Weight Drill Pipe

Primary

HIGH RISK

Accumulated across 4 reports: Heavy Weight Drill Pipe (total joints: 4.0, total length_ft: 1585.04). Risk from usage proxy and well NPT.

95%

Risk Score

Operating Hours

250 / 300 hrs

Action: Inspect

Next maintenance: 50 hours

Hydraulic Jar

Primary

MEDIUM RISK

Accumulated across 2 reports: Hydraulic Jar (total joints: 2.0, total length_ft: 63.7). Risk from usage proxy and well NPT.

54%

Risk Score

Operating Hours

327 / 2000 hrs

Action: Monitor

Next maintenance: 1673 hours

Summary Reports

6.1 Summary Reports page

- Click on “Summary Reports” on the dashboard page in the layout section, to see the summary details of all wells in your account.

DrillMain Intelligence

Wells

Summary Reports

Well Location

AI Assistant

Log out

Fleet-Wide Reports

Comprehensive analysis across all wells

CSVPDFFull Report (JSON)

Active Wells

1

Total NPT

30.0 hrs

Total Events

124

Critical Events

10

Avg Productivity

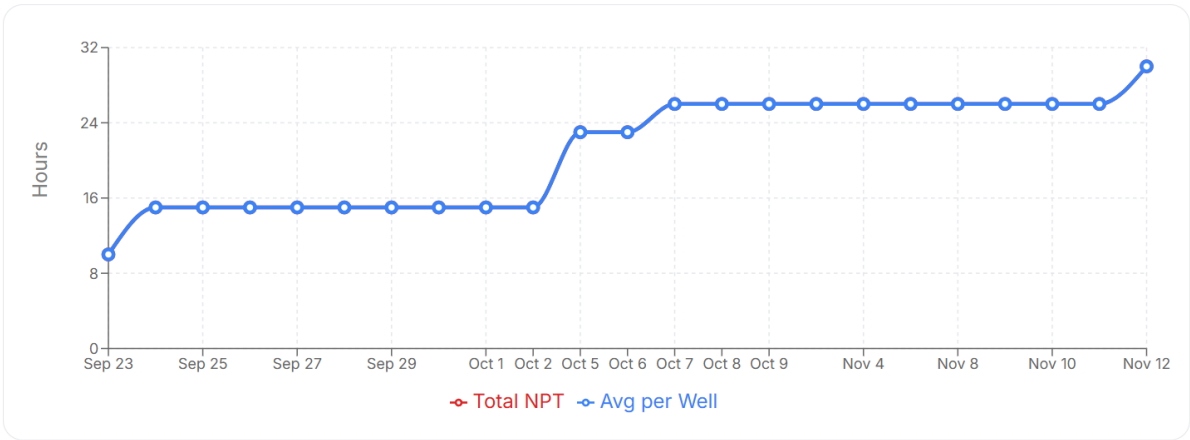
94.6%

NPT Comparison Across Wells

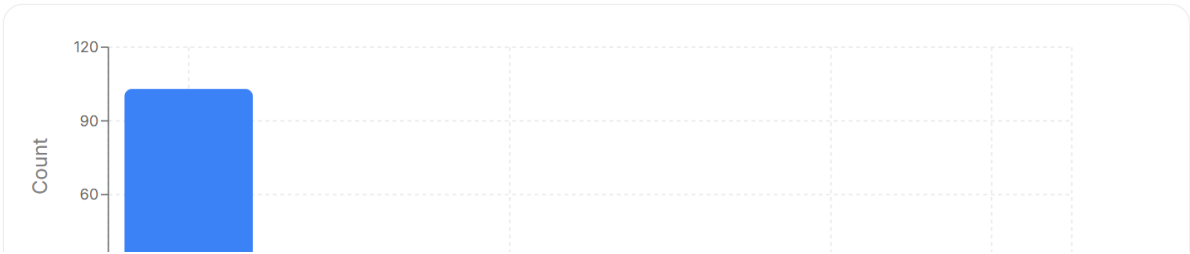
NPT Hours

Well-03

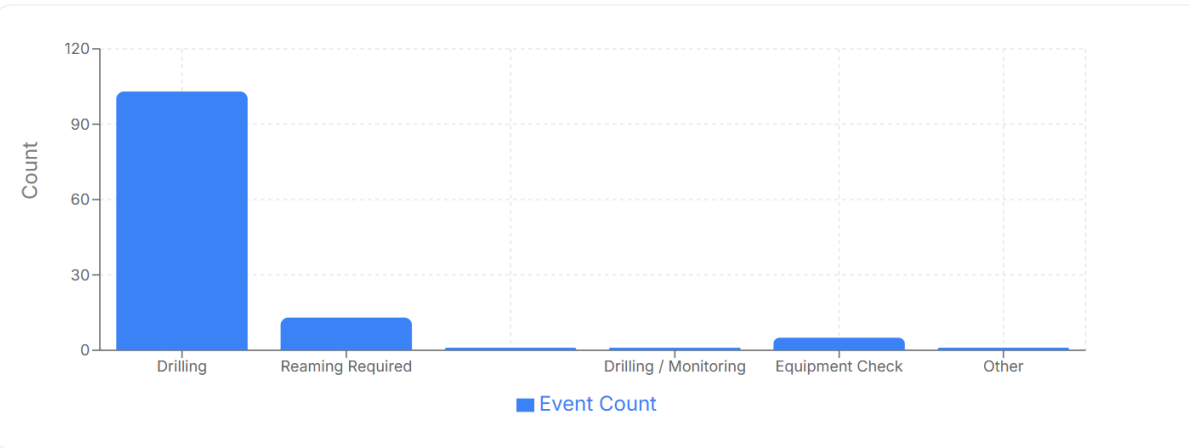
Fleet NPT Trend (Cumulative)



Event Type Distribution (All Wells)



Event Type Distribution (All Wells)



Events per Well



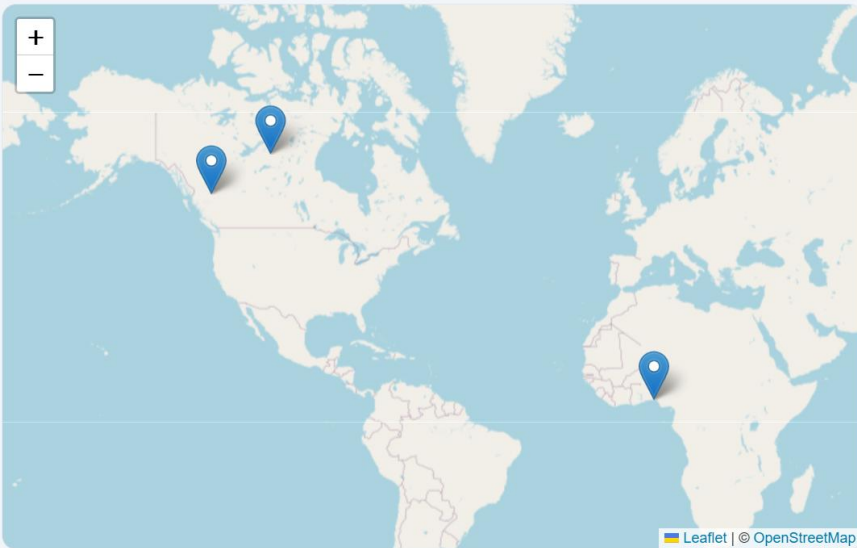
Location

7.1 Well Location page

- Click on “Well Location” on the dashboard page in the layout section, to see the location of all wells in your account on a map.

Well Location

Map overview of well locations. Wells are placed by geocoding their location or name.



Wells on map (3)

NNPC
LAGOS

SHELL
Canada

Shell
BC